

FINAL PROJECT REPORT

Metro Ticket Booking System in ServiceNow

Automated Fare Calculation, Digital QR Ticket Generation and Request Processing

Metro Ticket Booking - ServiceNow Workflow

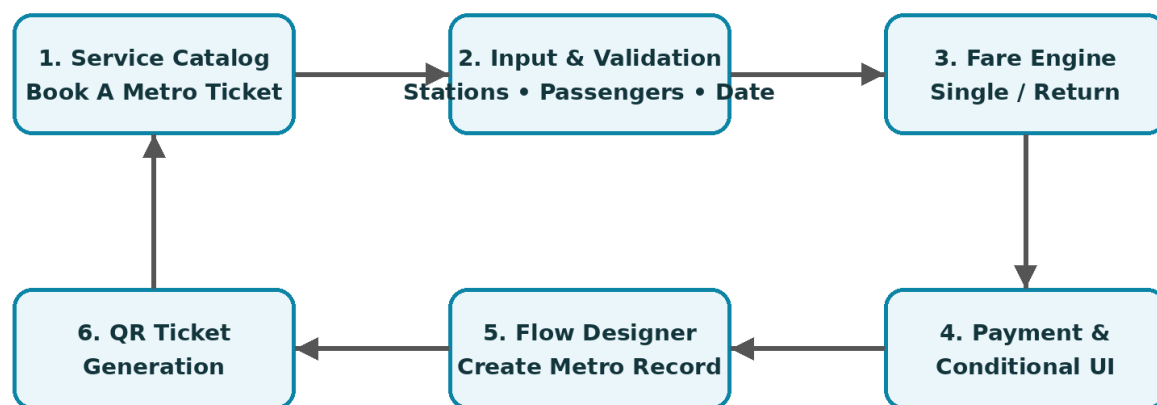


Figure 1. End-to-end solution workflow implemented in ServiceNow.

Platform: ServiceNow

Application Area: Service Catalog / Flow Designer / Client Scripts / Custom Data Table

Final validation evidence: September 2026

Document Control

Item	Details
Project Title	Metro Ticket Booking System in ServiceNow
Primary ServiceNow Component	Catalog Item – Book A Metro Ticket
Core Features	Station selection, passenger validation, fare calculation, payment selection, travel date, QR ticket generation, request/order processing
Automation	Catalog Client Scripts, Catalog UI Policy, Flow Designer
Data Layer	Metro Station's Details table and Metro Database record creation flow
Final Tested Scenario	10 passengers; Ashok Nagar → Chennai Central; Return journey; Card payment; Travel date 2026-09-25
Fare Result	Single Journey Fare ₹400 + Return Fare ₹400 = Total Fare ₹800

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1. Executive Summary

The Metro Ticket Booking System is a ServiceNow-based digital ticketing solution designed to move a metro ticket booking process from manual fare calculation and physical ticket handling to a guided, automated Service Catalog experience.

The implementation combines catalog variables, station reference data, client-side validation, dynamic fare calculation, conditional payment fields, Flow Designer automation, and digital QR ticket generation.

The final validation scenario confirms the required fare behavior: for 10 passengers, the Single Journey Fare is ₹400 and the Return Fare is ₹400, producing a Total Fare of ₹800.

The project reference material describes a phased implementation covering requirement analysis, backend/data configuration, UI/UX customization, testing/security, and deployment/documentation. This report consolidates those implementation details with the latest output evidence.

2. Introduction and Problem Statement

Traditional ticket booking can involve station queues, manual fare lookup, repetitive data entry and physical ticket handling. These activities create avoidable operational effort and increase the possibility of inconsistent fare calculations or incomplete booking information.

The purpose of this project is to demonstrate how ServiceNow can be used as a digital service platform for a structured metro-ticket booking workflow. The Service Catalog item is the user-facing transaction form, while ServiceNow configuration and scripting capabilities automate the logic behind that form.

3. Objectives

Digitize ticket booking using a guided Service Catalog experience.

Automate fare calculation from configured route, passenger and journey inputs.

Support both Single and Return journey types with separate fare components.

Improve data quality using station references, mandatory controls and validation.

Provide digital QR-based ticket verification.

Persist submitted booking information through Flow Designer.

Maintain station information in a dedicated master table for easier administration.

4. Requirements Specification

Functional requirements include station selection, passenger entry, Single/Return selection, automatic fare calculation, payment selection, conditional payment detail, travel date, QR generation, request processing and backend record creation.

Non-functional requirements include usability, consistency, maintainability, traceability, data integrity and extensibility.

5. System Scope and Stakeholders

In scope are the Service Catalog booking interface, station reference data, passenger validation, fare logic, payment controls, travel date, Flow Designer processing, QR output and order confirmation.

Primary stakeholders are passengers/end users, metro operations teams, station/operations administrators, ServiceNow administrators/developers and QA/support personnel.

6. Solution Architecture

The architecture separates the user-facing catalog experience from the configuration/data and automation layers. The catalog is the presentation layer; client scripts and UI policies implement behavior; custom tables provide data; Flow Designer persists the transaction; QR and order views provide verification.

7. ServiceNow Data Model

A dedicated custom table, Metro Station's Details, is used as the reference source for station selection. The configuration evidence shows Station Name, Station code and Station Status fields. The station table provides reusable reference records for the catalog form.

8. Catalog Item and Variable Design

The primary transaction is implemented as the Book A Metro Ticket catalog item. The configured variables cover Starting From, Going to, No of Passengers, Type of Journey, Single Journey Fare, Return Fare, Mode of Payment, conditional payment mode, Travel Date and Total Fare.

9. Automation and Business Logic

Catalog Client Scripts are used for fare calculation, passenger-count validation, QR generation and payment feedback.

The fare logic reads origin, destination, journey type and passenger count, matches configured route rules and writes the appropriate fare values.

The passenger validation logic checks that the passenger count is numeric before it is used for calculation.

The QR-generation logic is initiated on submission and presents a digital ticket using transaction/request-related data.

10. Fare Calculation Rules

The final business requirement is explicit: for 10 passengers, the Single Journey Fare is ₹400 and the Return Fare is ₹400. The Total Fare is therefore $₹400 + ₹400 = ₹800$.

The Return journey is treated as two fare components rather than replacing the Single Journey Fare value with a single combined number.

11. QR Ticket Generation

The QR output provides a scannable digital ticket containing a QR code, ticket number, fare information and travel guidance. The supplied QR evidence shows ticket number RITM0010048 and a displayed total fare of ₹60.00 from a separate successful run.

12. Payment and Conditional UI

Payment choices are Cash, UPI, Card and Others. When Others is selected, a Catalog UI Policy reveals an additional payment-mode field. When Card is selected, the form provides a contextual readiness message.

13. Flow Designer and Record Processing

The flow uses a Service Catalog trigger, Get Catalog Variables action and Create Record action to transfer submitted catalog values into Metro Database. This separates user interaction from persistence and provides an extension point for notifications, approvals and reporting.

14. Validation and Error Handling

Mandatory fields prevent incomplete submissions. Reference fields reduce station-name errors. Passenger validation rejects non-numeric input. Journey-dependent logic clears incompatible fare fields. Conditional payment UI reduces irrelevant fields. For production, critical fare validation should also be enforced server-side.

15. Testing Strategy and Test Cases

Testing covered catalog loading, station references, passenger validation, Single and Return calculations, the 10-passenger ₹800 acceptance scenario, payment selection, conditional UI, travel date, QR generation, request processing and order confirmation.

16. Output and Result Evidence

The screenshots in this report provide implementation and output evidence from the completed ServiceNow configuration. They include the catalog item, station table, reference variable, station record, QR ticket and final order-status output.

17. Security, Data Integrity and Maintainability

Centralized reference data, mandatory controls, validation, calculated fields and controlled flow mapping improve data integrity. Production deployments should use role-based administration and server-side validation for critical transaction values.

18. Deployment and Operational Considerations

Use controlled application/update-set deployment, verify station reference data, validate script variable names, test QR connectivity, review currency settings and run regression tests after deployment.

19. Limitations and Currency Display Note

The final evidence shows the business fare as ₹800 while the ServiceNow Order Status line renders \$800.00. The calculation itself is consistent; the difference is a currency-display/catalog configuration issue that should be standardized before production use.

A separate QR output screenshot shows ₹60.00 and ticket number RITM0010048. It demonstrates the QR mechanism from another successful run and is not the same fare-validation scenario as the 10-passenger Return case.

20. Future Enhancements

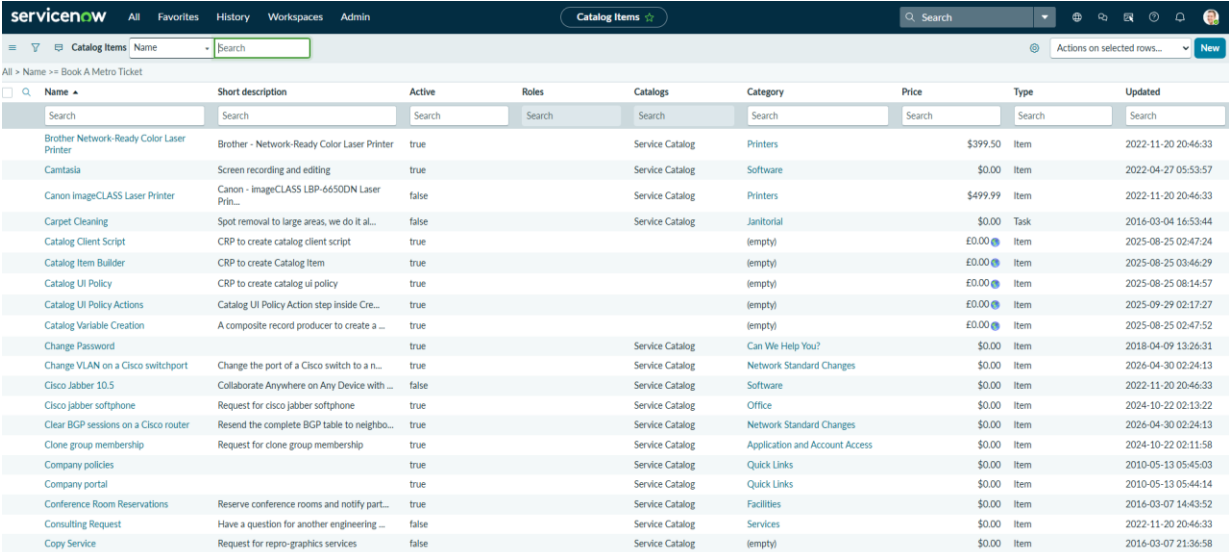
Introduce a normalized route/fare table instead of hard-coded route branches.
Add server-side fare validation before record creation.
Integrate production-grade QR verification with expiry and replay protection.
Add notifications, cancellation/refund flows and passenger categories.
Add dashboards for booking volume, route usage and revenue.
Add role-based administration and automated regression tests.
Standardize INR currency formatting across catalog, ticket and order-status views.

21. Conclusion

The completed solution demonstrates a complete ServiceNow transaction from booking input through automated fare calculation, payment selection, backend processing and digital QR-ticket output.

The key acceptance rule has been validated: for 10 passengers on a Return journey, Single Journey Fare ₹400 + Return Fare ₹400 = Total Fare ₹800.

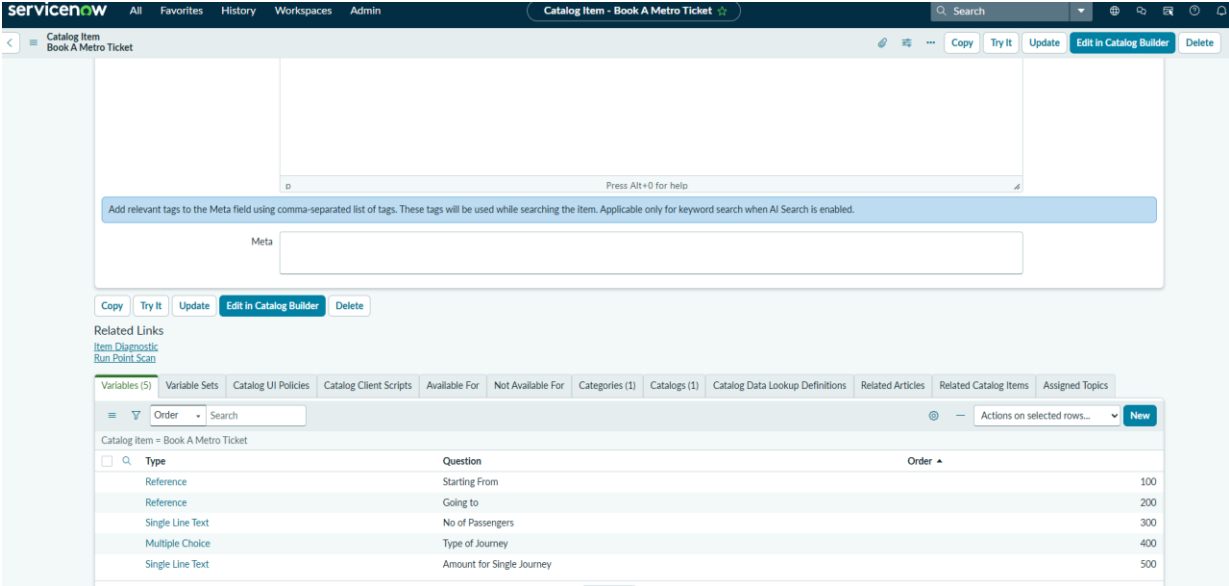
22. Output and Result Evidence



The screenshot shows the ServiceNow Catalog Items list. The table has columns for Name, Short description, Active, Roles, Catalogs, Category, Price, Type, and Updated. The list includes various items such as Brother Network-Ready Color Laser Printer, Camtasia, Canon ImageCLASS Laser Printer, Carpet Cleaning, Catalog Client Script, Catalog Item Builder, Catalog UI Policy, Catalog UI Policy Actions, Catalog Variable Creation, Change Password, Change VLAN on a Cisco switchport, Cisco Jabber 10.5, Cisco Jabber softphone, Clear BGP sessions on a Cisco router, Clone group membership, Company policies, Company portal, Conference Room Reservations, Consulting Request, and Copy Service.

Name	Short description	Active	Roles	Catalogs	Category	Price	Type	Updated
Brother Network-Ready Color Laser Printer	Brother - Network-Ready Color Laser Printer	true		Service Catalog	Printers	\$399.50	Item	2022-11-20 20:46:33
Camtasia	Screen recording and editing	true		Service Catalog	Software	\$0.00	Item	2022-04-27 05:53:57
Canon ImageCLASS Laser Printer	Canon - imageCLASS LBP-6650DN Laser Prin...	false		Service Catalog	Printers	\$499.99	Item	2022-11-20 20:46:33
Carpet Cleaning	Spot removal to large areas, we do it al...	false		Service Catalog	Janitorial	\$0.00	Task	2016-03-04 16:53:44
Catalog Client Script	CRP to create catalog client script	true			(empty)	£0.00	Item	2025-08-25 02:47:24
Catalog Item Builder	CRP to create Catalog Item	true			(empty)	£0.00	Item	2025-08-25 03:46:29
Catalog UI Policy	CRP to create catalog ui policy	true			(empty)	£0.00	Item	2025-08-25 08:14:57
Catalog UI Policy Actions	Catalog UI Policy Action step inside Cre...	true			(empty)	£0.00	Item	2025-09-29 02:17:27
Catalog Variable Creation	A composite record producer to create a ...	true			(empty)	£0.00	Item	2025-08-25 02:47:52
Change Password		true		Service Catalog	Can We Help You?	\$0.00	Item	2018-04-09 13:26:31
Change VLAN on a Cisco switchport	Change the port of a Cisco switch to a n...	true		Service Catalog	Network Standard Changes	\$0.00	Item	2026-04-30 02:24:13
Cisco Jabber 10.5	Collaborate Anywhere on Any Device with ...	false		Service Catalog	Software	\$0.00	Item	2022-11-20 20:46:33
Cisco Jabber softphone	Request for cisco jabber softphone	true		Service Catalog	Office	\$0.00	Item	2024-10-22 02:13:22
Clear BGP sessions on a Cisco router	Resend the complete BGP table to neighbo...	true		Service Catalog	Network Standard Changes	\$0.00	Item	2026-04-30 02:24:13
Clone group membership	Request for clone group membership	true		Service Catalog	Application and Account Access	\$0.00	Item	2024-10-22 02:11:58
Company policies		true		Service Catalog	Quick Links	\$0.00	Item	2010-05-13 05:45:03
Company portal		true		Service Catalog	Quick Links	\$0.00	Item	2010-05-13 05:44:14
Conference Room Reservations	Reserve conference rooms and notify part...	true		Service Catalog	Facilities	\$0.00	Item	2016-03-07 14:43:52
Consulting Request	Have a question for another engineering ...	false		Service Catalog	Services	\$0.00	Item	2022-11-20 20:46:33
Copy Service	Request for repro-graphics services	false		Service Catalog	(empty)	\$0.00	Item	2016-03-07 21:36:58

Figure 2. ServiceNow Catalog Items list used during catalog-item configuration.



The screenshot shows the configuration page for the 'Book A Metro Ticket' catalog item. It includes a 'Meta' field with a placeholder for relevant tags. Below this, there are tabs for 'Variables (5)', 'Variable Sets', 'Catalog UI Policies', 'Catalog Client Scripts', 'Available For', 'Not Available For', 'Categories (1)', 'Catalogs (1)', 'Catalog Data Lookup Definitions', 'Related Articles', 'Related Catalog Items', and 'Assigned Topics'. The 'Variables (5)' tab is active, showing a table of variables.

Type	Question	Order
Reference	Starting From	100
Reference	Going to	200
Single Line Text	No of Passengers	300
Multiple Choice	Type of Journey	400
Single Line Text	Amount for Single Journey	500

Figure 3. Book A Metro Ticket item and configured variables.

Table
Metro Station's Details

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label **Metro Station's Details** Application Global

* Name **u_metro_station_s_details** Remote Table

Columns Controls Application Access

Table Columns for text Search 1 to 9 of 9 New

Column label	Type	Reference	Max length	Default value	Display
Updated	Date/Time	(empty)	40	40	false
Station Status	Choice	(empty)	40	40	false
Created	Date/Time	(empty)	40	40	false
Station code	String	(empty)	30	30	false
Station Name	String	(empty)	100	100	false
Sys ID	Sys ID (GUID)	(empty)	32	32	false
Created by	String	(empty)	40	40	false
Previous Created by	Integer	(empty)	40	40	false
Updated by	String	(empty)	40	40	false

Delete Update Delete All Records

Related Links
Form Builder
Design Form

Figure 4. Metro Station's Details table and station master fields.

servicenow All Favorites History Workspaces Admin Variable - Starting From

Variable Starting From

Application Global Type Reference Catalog Item Book A Metro Ticket Order 100

Active Mandatory Read only Hidden Disable automatic slot fill based on user context

Question Annotation **Type Specifications** Default Value Auto-populate Permission Availability

Values specific to this variable **Type**

Variable Width System Default Width (50) % Not honored in 2 column container

* Reference Metro Station's Details [u_metro_station_s_details]

Use reference qualifier Simple

Reference qualifier condition Add Filter Condition Add OR Clause

-- choose field -- -- oper -- -- value --

Some variable types support specific **Variable Attributes**

Variable attributes

Copy Update Delete

Related Links
Run Point Scan

Figure 5. Starting From reference variable configuration.

Metro Station's Details
New record

* Station Name Ameerpet

* Station code AMR

* Station Status -- None --

Submit

Figure 6. Station record data-entry example.

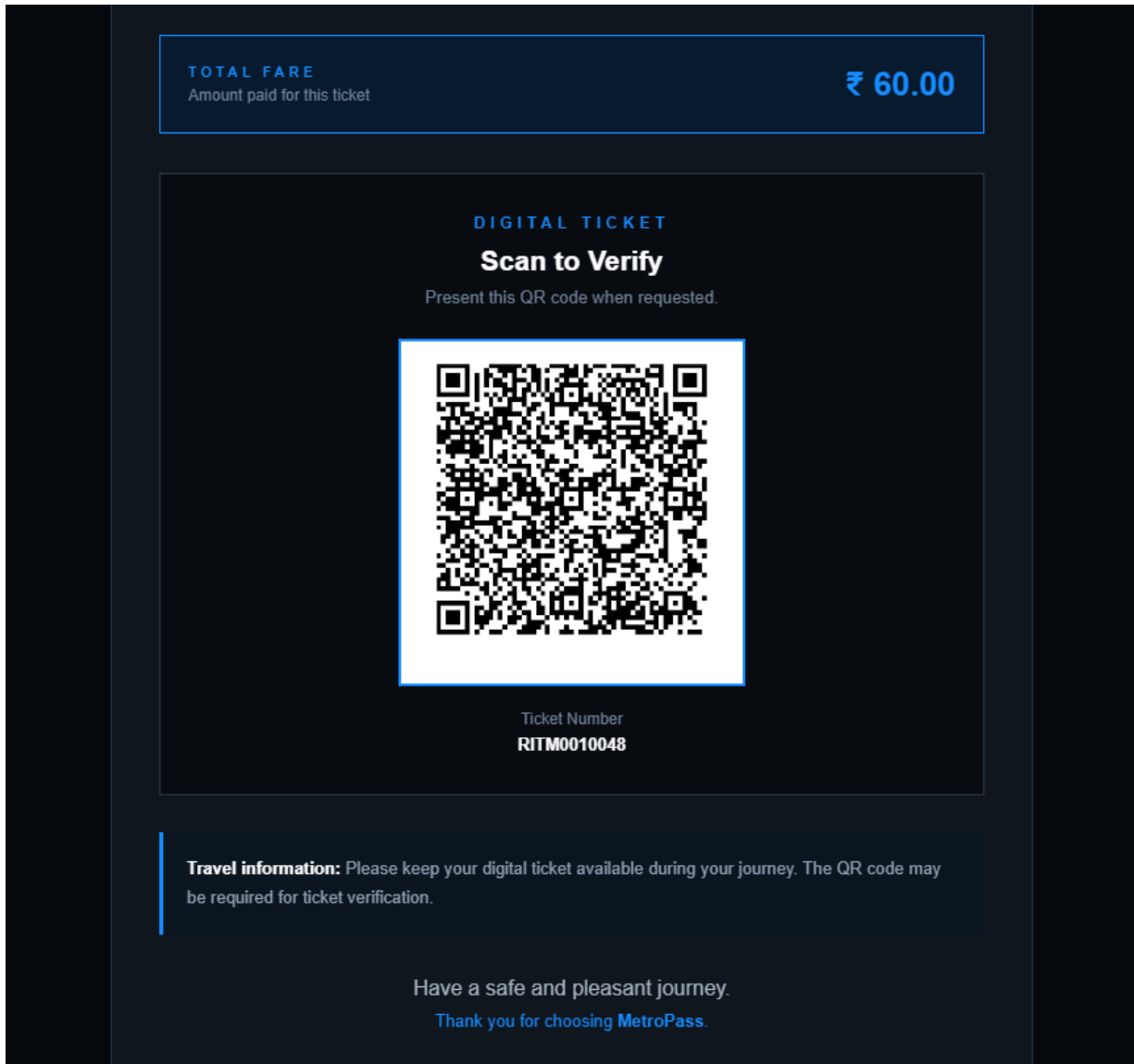


Figure 7. Generated digital QR ticket.

Order Status						Back to Catalog	Continue Shopping	Home
Thank you, your request has been submitted								
Order Placed: 2026-09-22 05:05:21 Request Number: REQ0010048 ☆ Estimated Delivery Date of Complete Order: 2026-09-22								
Description	Delivery Date	Stage	Price (ea.)	Quantity	Total			
Book metro tickets with automatic fare calculation and QR ticket generation	2026-09-22	▶	\$800.00	1	\$800.00			
				Total	\$800.00			
Back to Catalog Continue Shopping						Home		

Figure 8. Final ServiceNow Order Status output for the ₹800 test booking.

23. Appendix – Configuration Summary

Component	Configured Role
Catalog Item	Book A Metro Ticket
Custom Table	Metro Station's Details
Booking Storage	Metro Database
Catalog Client Scripts	Fare calculation; passenger validation; QR generation; payment feedback
Catalog UI Policy	Show Enter payment mode when Mode of Payment = Others
Flow Designer	Service Catalog trigger → Get Catalog Variables → Create Record
Digital Output	QR ticket page / ticket presentation
Confirmation	ServiceNow Order Status / Request and RITM identifiers

24. Final Acceptance Scenario

Input / Output	Final Value
Starting From	Ashok Nagar
Going to	Chennai Central
No of Passengers	10
Type of Journey	Return
Single Journey Fare	₹400
Return Fare	₹400
Total Fare	₹800
Mode of Payment	Card
Travel Date	2026-09-25
ServiceNow Request	REQ0010048

Source and evidence note: This report consolidates the uploaded reference project document, implementation/configuration evidence and final validation screenshots supplied with the project.